

D3Net-based All-in-One Restoration for Smartphone Images with Dirty Lenses

SIDL adaptation, paper-baseline comparison, patch-size ablation, and failure analysis

MAIN RESULT 23.86 dB / 0.8419 Current best validation checkpoint: 256 crop, epoch 73	COMPARISON SCOPE AirNet / DiffUIR / D3Net SIDL paper test results vs. our validation results	BEST SETTING 256×256 crop Continued 256 training improved both PSNR and SSIM	MAIN BOTTLENECK Hard Dust / Mixed Hard split improved, but severe local and mixed contamination remains difficult
--	--	--	---

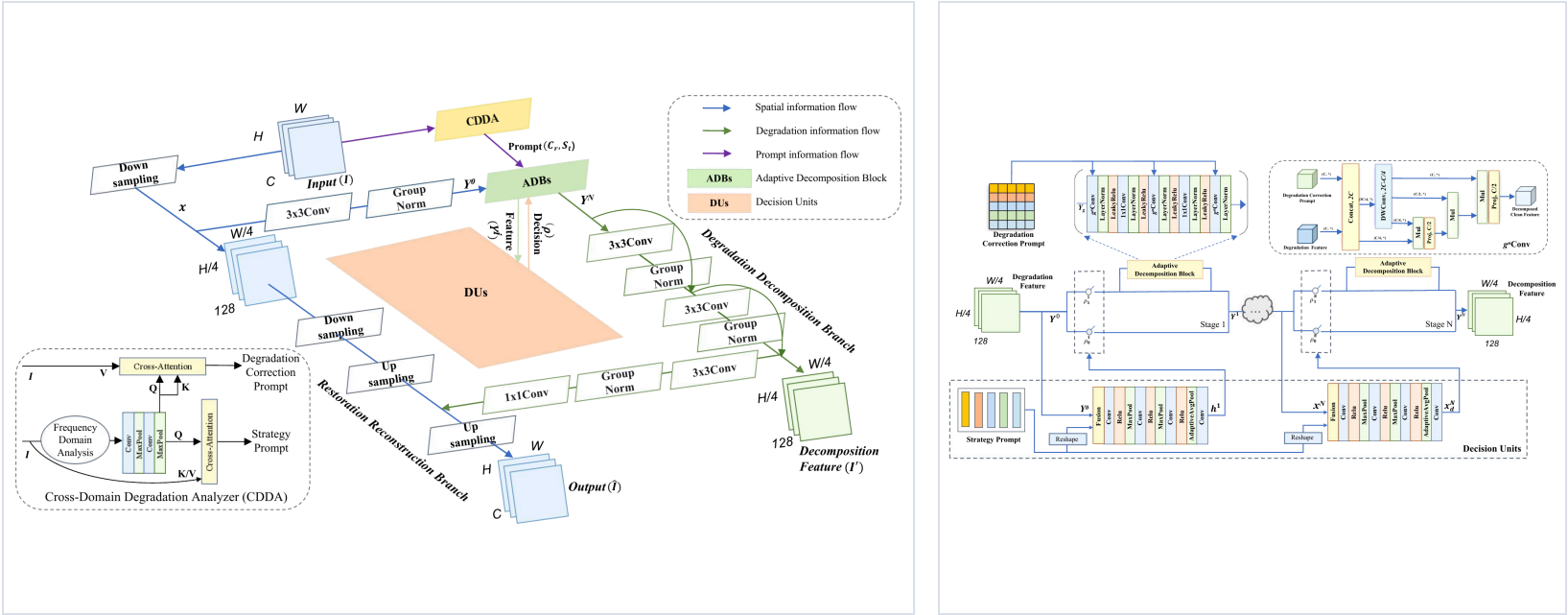
1. Problem & Dataset

SIDL is a real-world paired dataset for restoring smartphone images captured through contaminated lenses. The task is different from ordinary denoising or deblurring because lens contamination produces both local artifacts and global contrast degradation.

clean	dust	finger
water	scratch	mixed

Data 512×512 patchified degraded-clean pairs
Validation Type × difficulty split, PSNR / SSIM
Goal Analyze D3Net adaptation, not claim SIDL SOTA

2. Method: D3Net Adaptation



D3Net predicts a residual correction: restored image = input + residual. Frequency-domain cues and dynamic decomposition blocks allow degradation-aware processing for mixed and local contamination patterns.

4. SIDL Paper Baseline Comparison

AirNet and DiffUIR are SIDL paper test-set values. D3Net is our epoch-73 validation checkpoint, so this is task context rather than a strict leaderboard comparison.

METHOD	LEVEL	CLEAN	DUST	FINGER	WATER	SCRATCH	MIXED	AVG.
AirNet	Easy	27.10/0.9329	23.40/0.8669	25.10/0.8708	24.85/0.8658	25.84/0.8917	25.30/0.8422	25.26/0.8784
	Medium	26.13/0.8899	19.69/0.7577	22.53/0.8191	20.36/0.7656	21.24/0.8051	16.79/0.7542	21.12/0.7986
	Hard	23.79/0.8527	16.18/0.7350	15.83/0.6432	16.64/0.7449	18.13/0.7975	14.48/0.7047	17.50/0.7463
DiffUIR	Easy	34.51/0.9785	26.96/0.9276	28.82/0.9331	27.42/0.9220	30.14/0.9375	28.22/0.9021	29.34/0.9335
	Medium	33.25/0.9375	22.11/0.8325	26.16/0.8856	24.65/0.8647	27.07/0.9105	20.32/0.8368	25.36/0.8786
	Hard	29.47/0.9366	20.38/0.8602	18.93/0.7763	19.91/0.8500	21.27/0.8599	18.97/0.8409	21.71/0.8533
D3Net ours	Easy	32.78/0.9542	25.07/0.8859	24.59/0.7733	26.38/0.9196	27.84/0.8980	23.32/0.8552	26.66/0.8810
	Medium	30.26/0.9424	24.03/0.8797	23.69/0.8467	24.05/0.8577	27.41/0.9126	22.56/0.8869	25.33/0.8877
	Hard	27.70/0.8502	17.05/0.7086	20.23/0.7698	20.22/0.7986	20.38/0.7477	17.98/0.7900	20.59/0.7775

5. Our Ablation

RUN	INIT.	CROP	EPOCH	PSNR	SSIM
128 baseline	scratch	128	34	23.33	0.8357
256 main	scratch	256	34	23.82	0.8398
256 continued	latest	256	73	23.86	0.8419
256 low-LR	256 best	256	3	23.76	0.8409
512 refinement	256 best	512	2	23.27	0.8336

128	23.33
256	23.82
256 cont.	23.86
256 polish	23.76
512	23.27

6. Failure Analysis

TYPE	AVG. PSNR	AVG. SSIM
Clean	29.87	0.9125
Dust	21.90	0.8217
Finger	21.97	0.7897
Water	22.87	0.8433
Scratch	25.08	0.8505
Mixed	20.67	0.8311

Hard Dust: **17.05 / 0.7086**
Hard Mixed: **17.98 / 0.7900**

Continued training improved Hard difficulty from 20.33 to 20.59 dB, while Easy PSNR dropped from 27.00 to 26.66 dB. This suggests a cross-difficulty trade-off.

7. Qualitative Results

Each panel shows input, D3Net output, and GT from the report assets.

1. finger / hard / Case051_F_001.png PSNR 19.28 dB SSIM 0.9316

Input Output GT

2. dust / medium / Case251_D_018.png PSNR 19.25 dB SSIM 0.7943

Input Output GT

3. scratch / hard / Case073_S_001.png PSNR 10.47 dB SSIM 0.6080

Input Output GT

4. water / hard / Case051_W_001.png PSNR 17.70 dB SSIM 0.8999

Input Output GT

5. mixed / medium / Case170_DS_028.png PSNR 21.41 dB SSIM 0.9282

Input Output GT

8. Takeaways

Current best Epoch 73 reaches 23.86 dB / 0.8419 on the validation split.	Trade-off observed Hard restoration improves, while Easy PSNR slightly decreases.	Future direction Loss-metric mismatch and difficulty-aware training remain open issues.
--	---	---